

Multiple Choice (10 marks)

1. How many gallons of a liquid that is 74 percent alcohol must be combined with 5 gallons of one that is 90 percent alcohol in order to obtain a mixture that is 84 percent alcohol?
 - (a) 3 gallons
 - (b) 5 gallons
 - (c) 8 gallons
 - (d) 2 gallons
 - (e) 1 gallon
2. Determine the molarity of a 40.0% solution of HCl which has a density of 1.20 g/ml.
 - (a) 14.15 M
 - (b) 16.20 M
 - (c) 15.00 M
 - (d) 12.75 M
 - (e) 9.95 M
3. What is the ionization constant, K_a , for a weak monoprotic acid if a 0.060 molar solution has a pH of 2.0?
 - (a) 3.0×10^{-3}
 - (a) 2.0×10^{-3}
 - (a) 1.7×10^{-1}
 - (a) 1.5×10^{-2}
 - (a) 2.0×10^{-1}
4. A gas at 250 K and 15 atm has a molar volume 12 per cent smaller than that calculated from the perfect gas law. Calculate the molar volume of the gas.
 - (a) $1.2 \text{ dm}^3 \text{ mol}^{-1}$
 - (b) $1.35 \text{ dm}^3 \text{ mol}^{-1}$
 - (c) $2.0 \text{ dm}^3 \text{ mol}^{-1}$
 - (d) $0.8 \text{ dm}^3 \text{ mol}^{-1}$
 - (e) $1.8 \text{ dm}^3 \text{ mol}^{-1}$

5. 2.3 g of ethanol ($\text{C}_2\text{H}_5\text{OH}$, molecular weight = 46 g/mole) is added to 500 g of water. Determine the molality of the resulting solution.
- (a) 0.12 molal
 - (b) 0.05 molal
 - (c) 0.4 molal
 - (d) 0.2 molal
 - (e) 0.1 molal

True / False (10 marks)

Answer True or False

- 6. True or False: The root mean square speed of oxygen (O_2) at the same temperature as hydrogen (H_2) is 400 m/sec.
- 7. True or False: The reaction of sulfuric acid with metallic calcium produces 33.6 liters of hydrogen gas at standard temperature and pressure (STP).
- 8. True or False: The pK_a of the weak monoprotic acid can be determined directly as equal to its pH of 4.26.
- 9. True or False: An alcohol contains no oxygen atoms in its functional group, making it a hydrocarbon.
- 10. True or False: In terms of reducing ability, the order is $\text{H}_2 > \text{Ni} > \text{Co} > \text{Al} > \text{Na} > \text{Ag}$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Given a household cleaning solution with a hydronium ion concentration of 10^{-11}M , *explain how to calculate*
- 11. Calculate the cell potential for the reaction $\text{Ni} + \text{Sn}^{2+} \rightarrow \text{Ni}^{2+} + \text{Sn}$, given the concentrations of Ni^{2+} and Sn^{2+} . Discuss the implications for the spontaneity of the reaction.

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